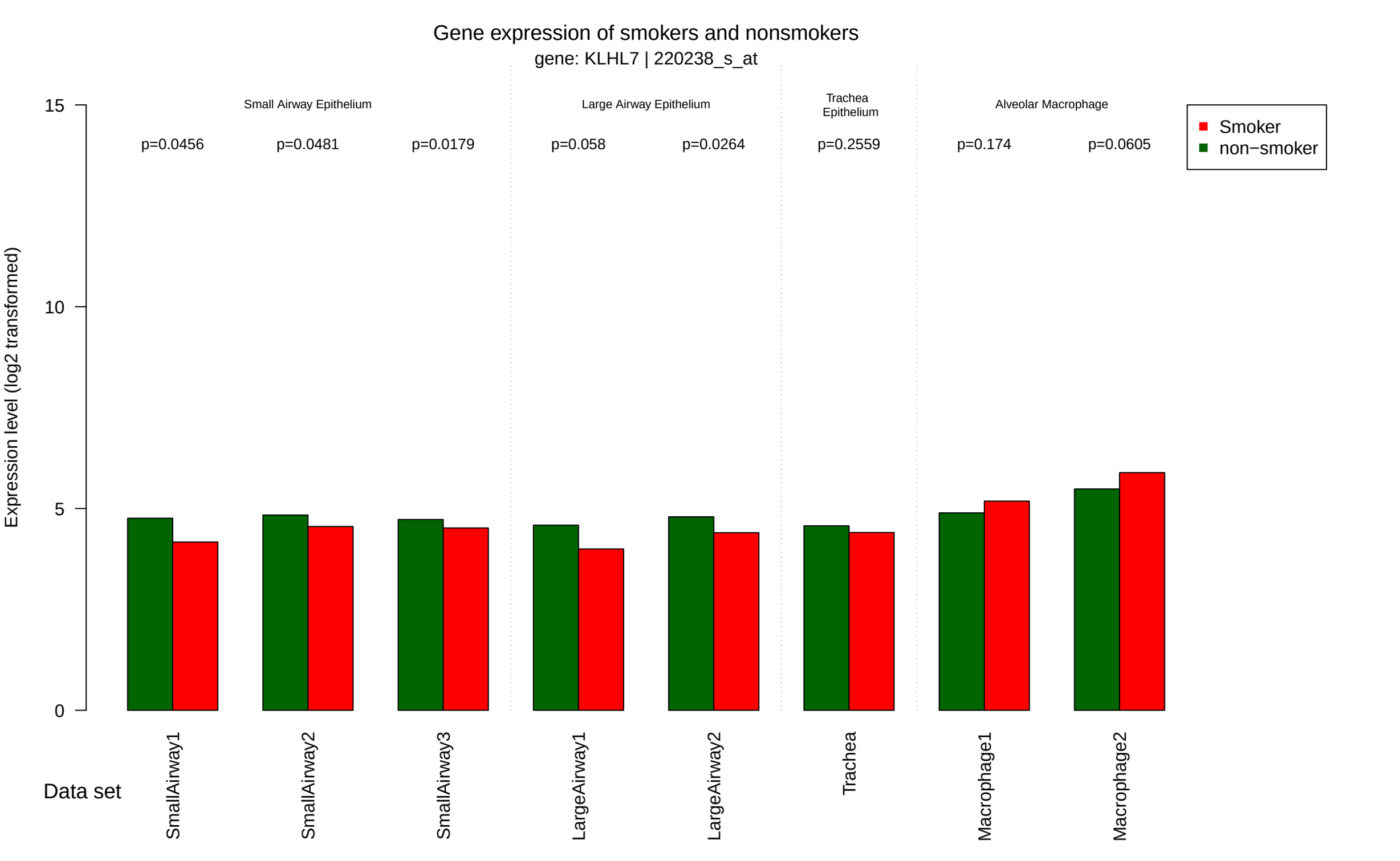
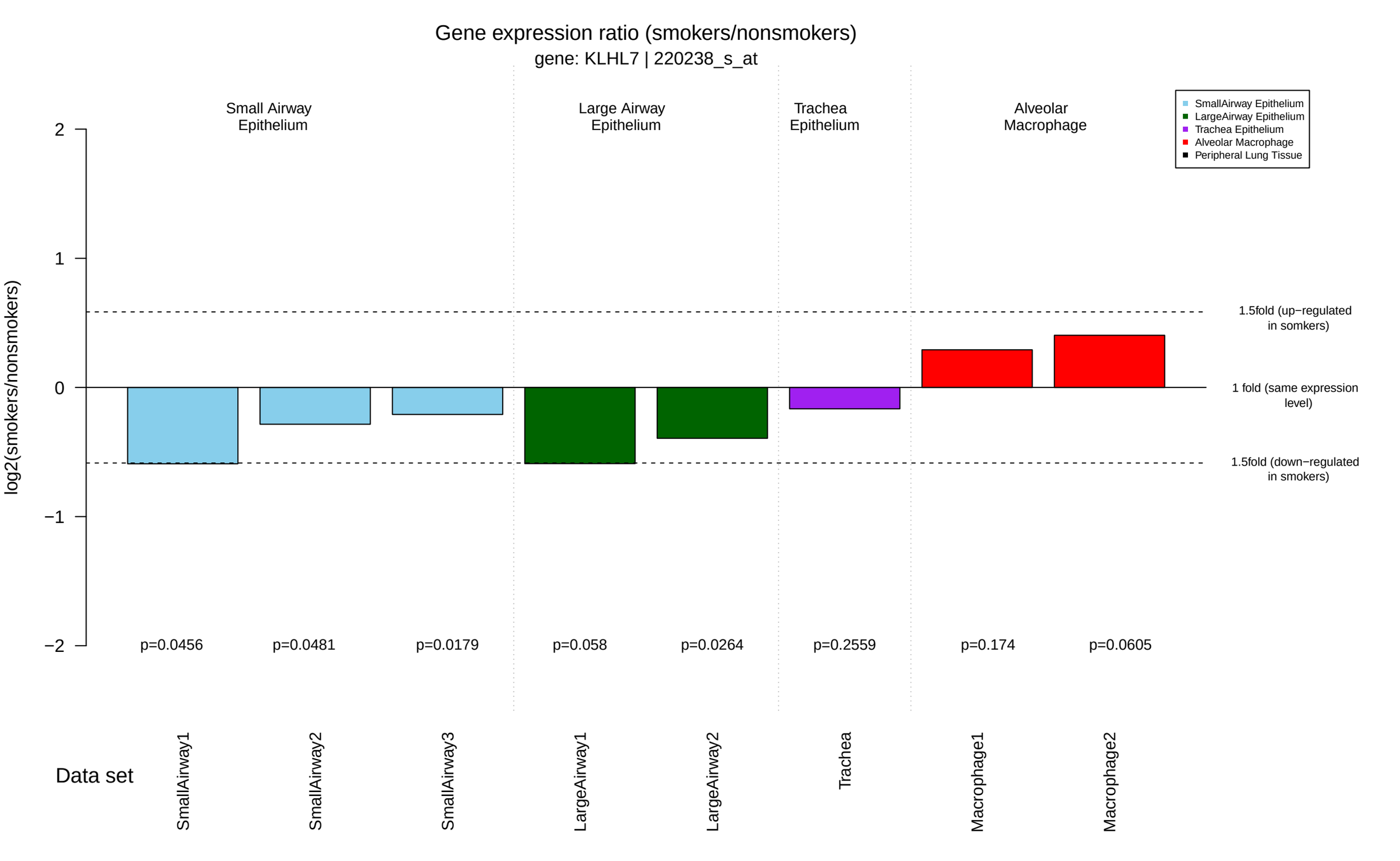
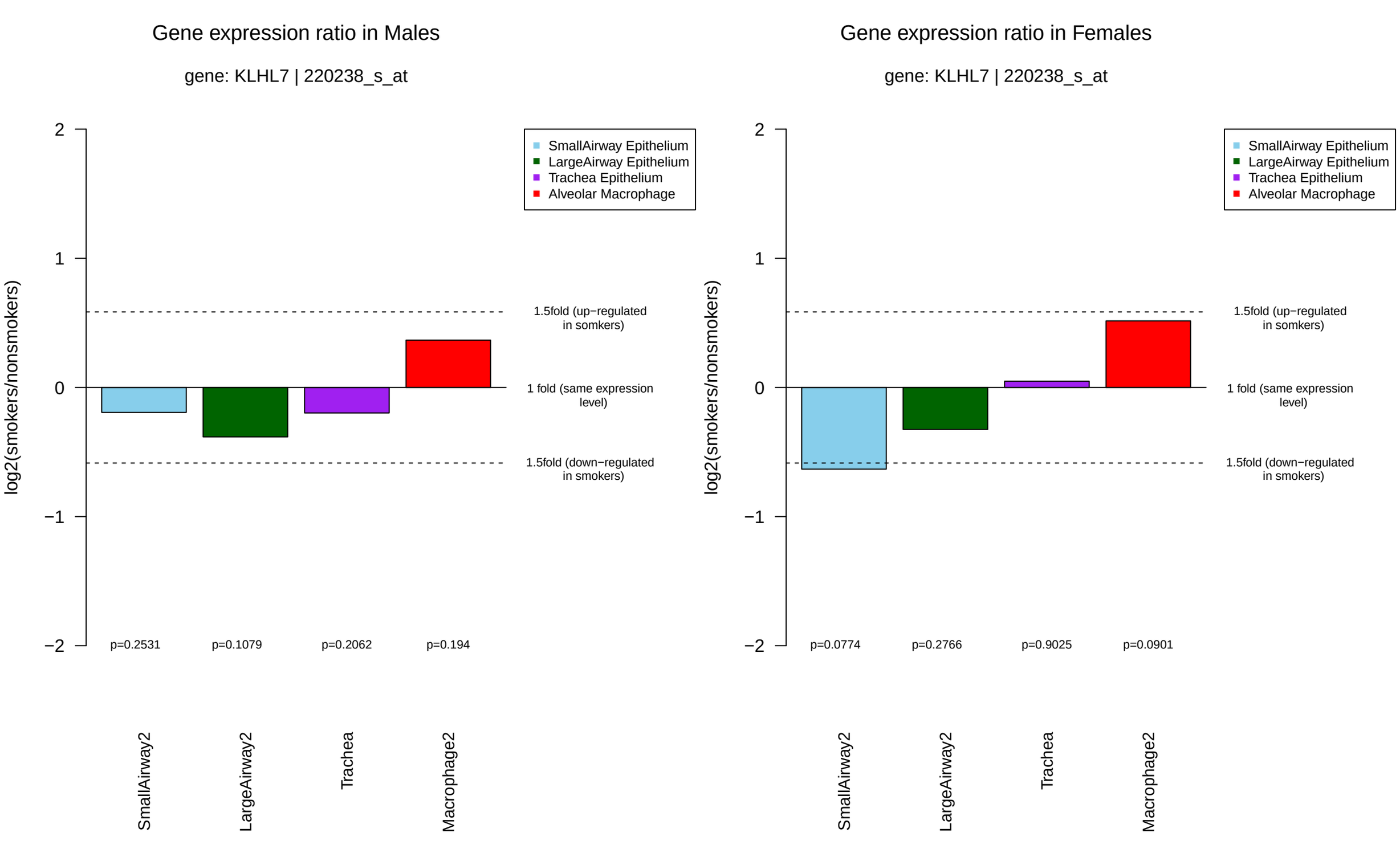
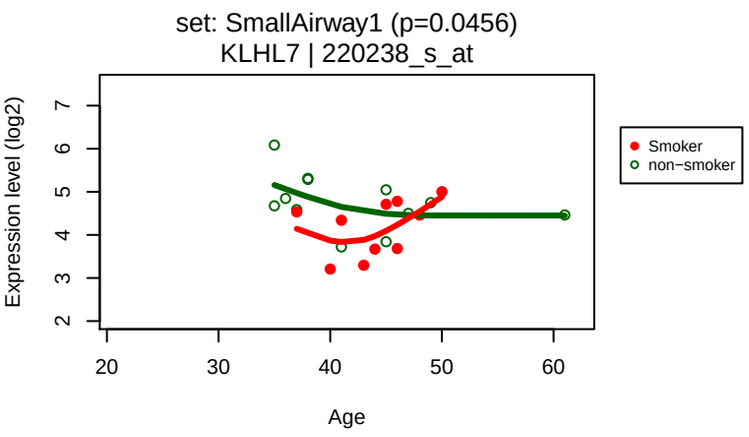
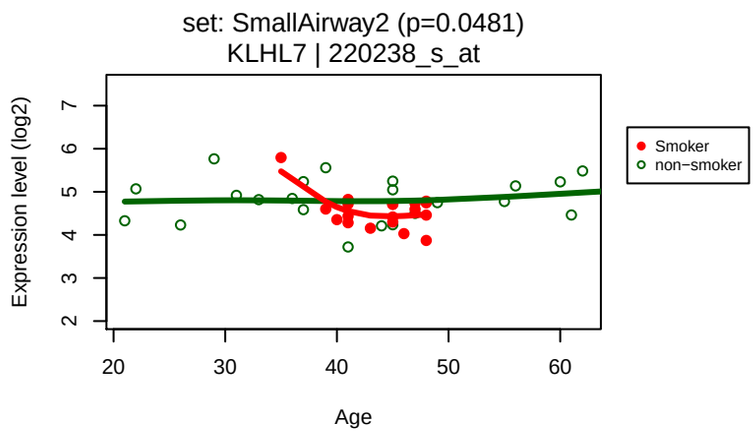
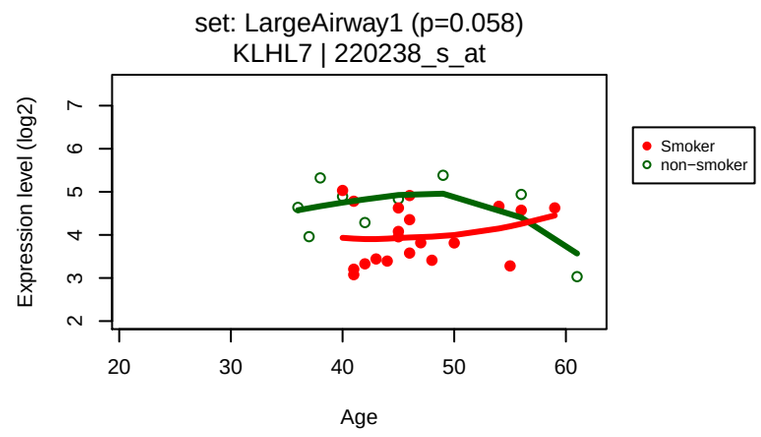

Gene expression of smokers and nonsmokers
gene: KLHL7 | 220238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0456
p=0.0481
p=0.0179
p=0.058
p=0.0264
p=0.2559
p=0.174
p=0.0605
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

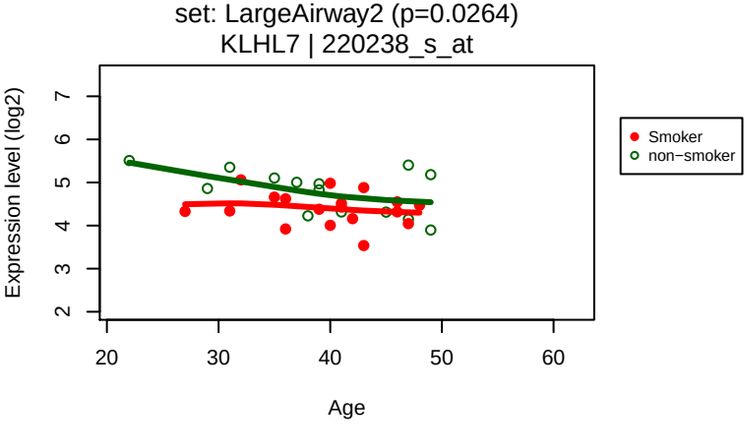

Gene expression ratio (smokers/nonsmokers)
gene: KLHL7 | 220238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0456
p=0.0481
p=0.0179
p=0.058
p=0.0264
p=0.2559
p=0.174
p=0.0605
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

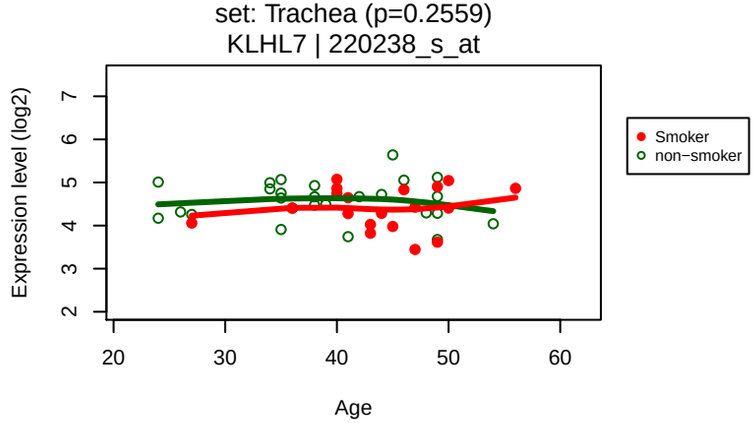

Gene expression ratio in Males
gene: KLHL7 | 220238_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2531
p=0.1079
p=0.2062
p=0.194
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KLHL7 | 220238_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0774
p=0.2766
p=0.9025
p=0.0901
SmallAirway2
LargeAirway2
Trachea
Macrophage2

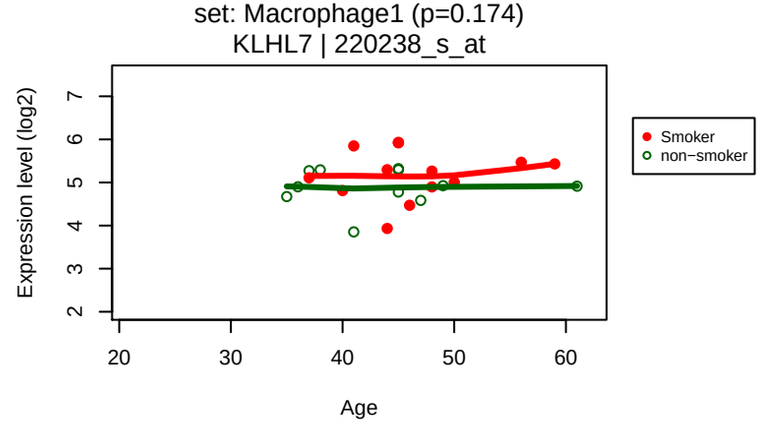

set: SmallAirway1 (p=0.0456)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker

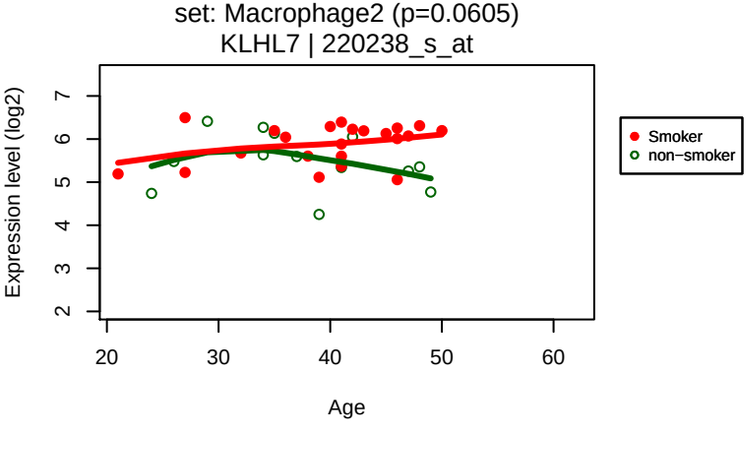

set: SmallAirway2 (p=0.0481)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker

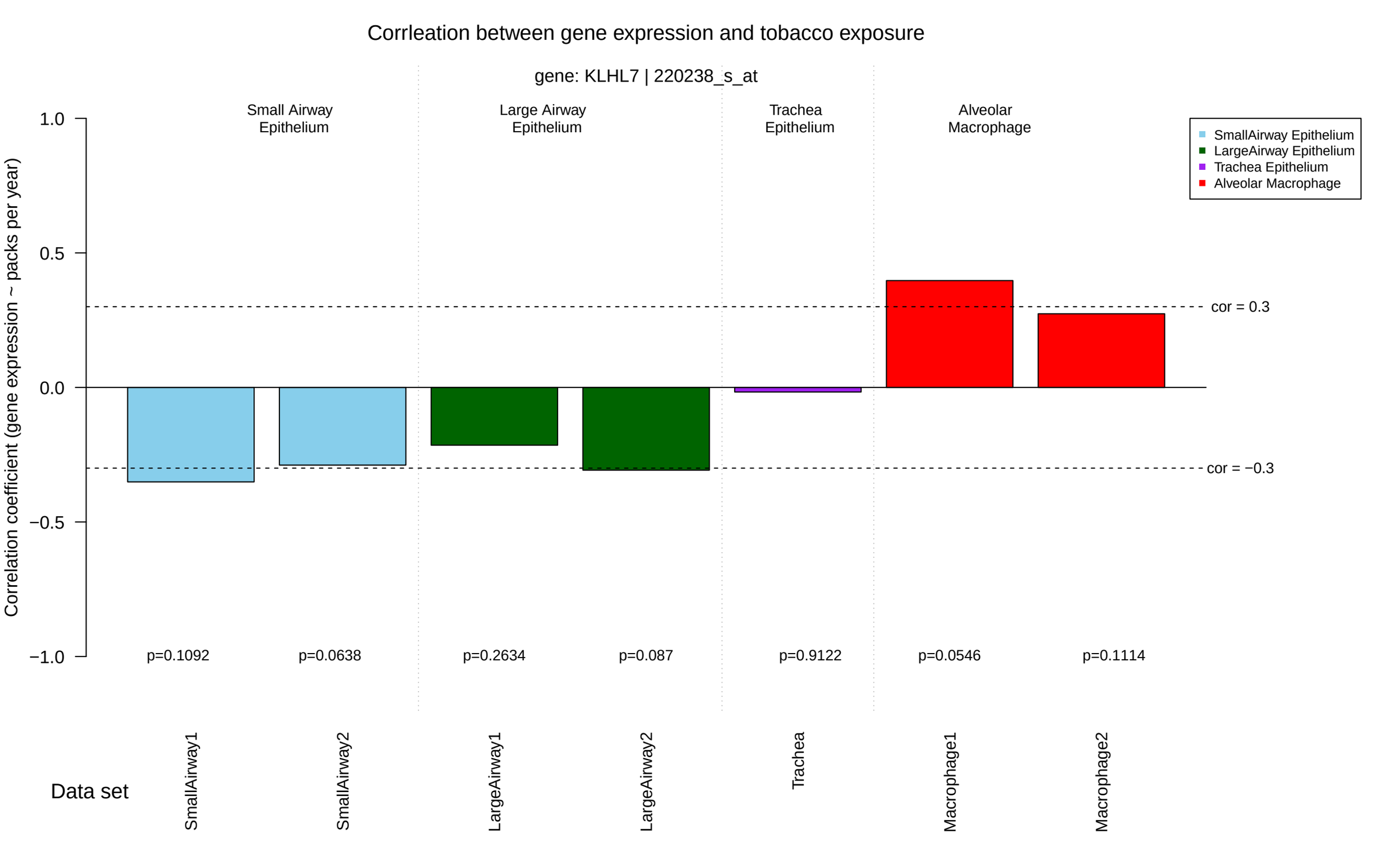

set: LargeAirway1 (p=0.058)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0264)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2559)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.174)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0605)
KLHL7 | 220238_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KLHL7 | 220238_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1092
p=0.0638
p=0.2634
p=0.087
p=0.9122
p=0.0546
p=0.1114
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2